



## Certificate of Analysis

Page 1 of 5

|                 |                                                                          |                          |                     |        |
|-----------------|--------------------------------------------------------------------------|--------------------------|---------------------|--------|
| <b>Client:</b>  | East Coast Drilling                                                      | <b>Lab No:</b>           | 2477371             | DWAPv1 |
| <b>Contact:</b> | East Coast Drilling<br>454 West Melton Road<br>RD 6<br>Christchurch 7676 | <b>Date Received:</b>    | 20-Nov-2020         |        |
|                 |                                                                          | <b>Date Reported:</b>    | 26-Nov-2020         |        |
|                 |                                                                          | <b>Quote No:</b>         | 99303               |        |
|                 |                                                                          | <b>Order No:</b>         |                     |        |
|                 |                                                                          | <b>Client Reference:</b> |                     |        |
|                 |                                                                          | <b>Submitted By:</b>     | East Coast Drilling |        |

### Sample Type: Aqueous

| Sample Name:                        |               | Haisma Lot 7 19-Nov-2020 5:00 pm | Guideline Value                     | Maximum Acceptable Values (MAV) |
|-------------------------------------|---------------|----------------------------------|-------------------------------------|---------------------------------|
| Lab Number:                         |               | 2477371.1                        |                                     |                                 |
| Individual Tests                    |               |                                  |                                     |                                 |
| Carbonate                           | g/m³ at 25°C  | < 1.0                            | -                                   | -                               |
| Bicarbonate                         | g/m³ at 25°C  | 60                               | -                                   | -                               |
| Langelier Saturation Index          |               | -0.8                             | -                                   | -                               |
| Calcium Hardness*                   | g/m³ as CaCO₃ | 35                               | -                                   | -                               |
| Sample Temperature*                 | °C            | 20.0                             | -                                   | -                               |
| Dissolved Calcium                   | g/m³          | 13.5                             | -                                   | -                               |
| Fluoride                            | g/m³          | 0.09                             | -                                   | 1.5                             |
| Total Ammoniacal-N                  | g/m³          | < 0.010                          | < 1.2                               | -                               |
| Total Phosphorus                    | g/m³          | 0.016                            | -                                   | -                               |
| Reactive Silica                     | g/m³ as SiO₂  | 21                               | -                                   | -                               |
| Non-Purgeable Organic Carbon (NPOC) | g/m³          | 0.5                              | -                                   | -                               |
| Routine Water Profile               |               |                                  |                                     |                                 |
| Turbidity                           | NTU           | 3.1                              | < 2.5                               | -                               |
| pH                                  | pH Units      | 7.7                              | 7.0 - 8.5                           | -                               |
| Total Alkalinity                    | g/m³ as CaCO₃ | 49                               | -                                   | -                               |
| Free Carbon Dioxide                 | g/m³ at 25°C  | 2.0                              | -                                   | -                               |
| Total Hardness                      | g/m³ as CaCO₃ | 59                               | < 200                               | -                               |
| Electrical Conductivity (EC)        | mS/m          | 17.6                             | -                                   | -                               |
| Electrical Conductivity (EC)        | µS/cm         | 176                              | -                                   | -                               |
| Approx Total Dissolved Salts        | g/m³          | 118                              | < 1000                              | -                               |
| Total Arsenic                       | g/m³          | < 0.0011                         | -                                   | 0.01                            |
| Total Boron                         | g/m³          | 0.0127                           | -                                   | 1.4                             |
| Total Calcium                       | g/m³          | 14.0                             | -                                   | -                               |
| Total Copper                        | g/m³          | < 0.00053                        | < 1                                 | 2                               |
| Total Iron                          | g/m³          | 0.166                            | < 0.2                               | -                               |
| Total Lead                          | g/m³          | 0.00012                          | -                                   | 0.01                            |
| Total Magnesium                     | g/m³          | 5.8                              | -                                   | -                               |
| Total Manganese                     | g/m³          | 0.0133                           | < 0.04 (Staining)<br>< 0.10 (Taste) | 0.4                             |
| Total Potassium                     | g/m³          | 0.99                             | -                                   | -                               |
| Total Sodium                        | g/m³          | 13.1                             | < 200                               | -                               |
| Total Zinc                          | g/m³          | < 0.0011                         | < 1.5                               | -                               |
| Chloride                            | g/m³          | 5.7                              | < 250                               | -                               |
| Nitrate-N                           | g/m³          | 6.2                              | -                                   | 11.3                            |
| Sulphate                            | g/m³          | 2.0                              | < 250                               | -                               |
| Total Coliforms and E.coli          |               |                                  |                                     |                                 |
| Total Coliforms                     | MPN / 100mL   | < 1                              | -                                   | -                               |
| Escherichia coli                    | MPN / 100mL   | < 1                              | -                                   | < 1                             |



This Laboratory is accredited by International Accreditation New Zealand (IANZ), which represents New Zealand in the International Laboratory Accreditation Cooperation (ILAC). Through the ILAC Mutual Recognition Arrangement (ILAC-MRA) this accreditation is internationally recognised. The tests reported herein have been performed in accordance with the terms of accreditation, with the exception of tests marked \* or any comments and interpretations, which are not accredited.

**Note:** The Guideline Values and Maximum Acceptable Values (MAV) are taken from the publication 'Drinking-water Standards for New Zealand 2005 (Revised 2018)', Ministry of Health. Copies of this publication are available from <https://www.health.govt.nz/publication/drinking-water-standards-new-zealand-2005-revised-2018>

The Maximum Acceptable Values (MAVs) have been defined by the Ministry of Health for parameters of health significance and should not be exceeded. The Guideline Values are the limits for aesthetic determinands that, if exceeded, may render the water unattractive to consumers.

Note that the units g/m<sup>3</sup> are the same as mg/L and ppm.

#### **pH/Alkalinity and Corrosiveness Assessment**

The pH of a water sample is a measure of its acidity or basicity. Waters with a low pH can be corrosive and those with a high pH can promote scale formation in pipes and hot water cylinders.

The guideline level for pH in drinking water is 7.0-8.5. Below this range the water will be corrosive and may cause problems with disinfection if such treatment is used.

The alkalinity of a water is a measure of its acid neutralising capacity and is usually related to the concentration of carbonate, bicarbonate and hydroxide. Low alkalinities (25 g/m<sup>3</sup>) promote corrosion and high alkalinities can cause problems with scale formation in metal pipes and tanks.

The pH of this water is within the NZ Drinking Water Guidelines, the ideal range being 7.0 to 8.0.

With the pH and alkalinity levels found, it is unlikely this water will be corrosive towards metal piping and fixtures.

#### **Hardness/Total Dissolved Salts Assessment**

The water contains a low amount of dissolved solids and would be regarded as being slightly hard.

#### **Nitrate Assessment**

Nitrate-nitrogen at elevated levels is considered undesirable in natural waters as this element can cause a health disorder called methaemaglobinaemia. Very young infants (less than six months old) are especially vulnerable. The Drinking-water Standards for New Zealand 2005 (Revised 2018) suggests a maximum permissible level of 11.3 g/m<sup>3</sup> as Nitrate-nitrogen (50 g/m<sup>3</sup> as Nitrate).

Nitrate-nitrogen was detected at a moderate level.

#### **Boron Assessment**

Boron may be present in natural waters and if present at high concentrations can be toxic to plants.

Boron was found at a low level in this water but would not give any cause for concern.

#### **Metals Assessment**

Iron and manganese are two problem elements that commonly occur in natural waters. These elements may cause unsightly stains and produce a brown/black precipitate. Iron is not toxic but manganese, at concentrations above 0.5 g/m<sup>3</sup>, may adversely affect health. At concentrations below this it may cause stains on clothing and sanitary ware.

Iron was found in this water at a low level.

Manganese was found in this water at a low level.

Treatment to remove iron and/or manganese should not be necessary.

#### **Bacteriological Tests**

The NZ Drinking Water Standards state that there should be no *Escherichia coli* (E coli) in water used for human consumption. The presence of these organisms would indicate that other pathogens of faecal origin may be present. Results obtained for Total Coliforms are only significant if the sample has not also been tested for E coli.

*Escherichia coli* was not detected in this sample.

#### **Final Assessment**

The parameter Turbidity did NOT meet the guidelines laid down in the publication 'Drinking-water Standards for New Zealand 2005 (Revised 2018)' published by the Ministry of Health for water which is suitable for drinking purposes.

## Summary of Methods

The following table(s) gives a brief description of the methods used to conduct the analyses for this job. The detection limits given below are those attainable in a relatively simple matrix. Detection limits may be higher for individual samples should insufficient sample be available, or if the matrix requires that dilutions be performed during analysis. A detection limit range indicates the lowest and highest detection limits in the associated suite of analytes. A full listing of compounds and detection limits are available from the laboratory upon request. Unless otherwise indicated, analyses were performed at Hill Laboratories, 28 Duke Street, Frankton, Hamilton 3204.

| Sample Type: Aqueous                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |           |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------|
| Test                                     | Method Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Default Detection Limit                   | Sample No |
| Individual Tests                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |           |
| Filtration, Unpreserved                  | Sample filtration through 0.45µm membrane filter. Performed at Hill Laboratories - Chemistry; 101c Waterloo Road, Christchurch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                         | 1         |
| Total Digestion                          | Nitric acid digestion. APHA 3030 E (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -                                         | 1         |
| Turbidity                                | Analysis using a Hach 2100 Turbidity meter. Analysed at Hill Laboratories - Chemistry; 101c Waterloo Road, Christchurch. APHA 2130 B 23 <sup>rd</sup> ed. 2017 (modified).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0.05 NTU                                  | 1         |
| pH                                       | pH meter. Analysed at Hill Laboratories - Chemistry; 101c Waterloo Road, Christchurch. APHA 4500-H <sup>+</sup> B 23 <sup>rd</sup> ed. 2017. Note: It is not possible to achieve the APHA Maximum Storage Recommendation for this test (15 min) when samples are analysed upon receipt at the laboratory, and not in the field. Samples and Standards are analysed at an equivalent laboratory temperature (typically 18 to 22 °C). Temperature compensation is used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.1 pH Units                              | 1         |
| Total Alkalinity                         | Titration to pH 4.5 (M-alkalinity), autotitrator. Analysed at Hill Laboratories - Chemistry; 101c Waterloo Road, Christchurch. APHA 2320 B (modified for Alkalinity <20) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.0 g/m <sup>3</sup> as CaCO <sub>3</sub> | 1         |
| Carbonate                                | Calculation: from alkalinity and pH, valid where TDS is not >500 mg/L and alkalinity is almost entirely due to hydroxides, carbonates or bicarbonates. APHA 4500-CO <sub>2</sub> D 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1.0 g/m <sup>3</sup> at 25°C              | 1         |
| Bicarbonate                              | Calculation: from alkalinity and pH, valid where TDS is not >500 mg/L and alkalinity is almost entirely due to hydroxides, carbonates or bicarbonates. APHA 4500-CO <sub>2</sub> D 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1.0 g/m <sup>3</sup> at 25°C              | 1         |
| Free Carbon Dioxide                      | Calculation: from alkalinity and pH, valid where TDS is not >500 mg/L and alkalinity is almost entirely due to hydroxides, carbonates or bicarbonates. APHA 4500-CO <sub>2</sub> D 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1.0 g/m <sup>3</sup> at 25°C              | 1         |
| Langelier Saturation Index               | Calculation: from pH, Total Alkalinity, Ionic Strength, Temperature* and Calcium. This calculation assumes that; 1) the calcium carbonate is in the form of calcite, 2) the effects associated with calcium ion pairs are negligible and 3) and the effect of alkalinity contributed or consumed by species other than HCO <sub>3</sub> <sup>-</sup> , CO <sub>3</sub> <sup>2-</sup> , OH <sup>-</sup> and H <sup>+</sup> is not significant.<br><br><b>*Note: For accurate calculation of the Langelier Saturation Index (LSI), the sample temperature should be taken using a calibrated thermometer at the time of sampling and recorded on the paperwork submitted with the sample. If a sample temperature is not supplied, a nominal temperature of 20°C will show in the results table above and be used in the calculation. In this case, please interpret the LSI result with caution.</b><br><br>APHA 2330 B 23 <sup>rd</sup> ed. 2017. | -                                         | 1         |
| Total Hardness                           | Calculation from Calcium and Magnesium. APHA 2340 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1.0 g/m <sup>3</sup> as CaCO <sub>3</sub> | 1         |
| Calcium Hardness*                        | Calculation from Total Calcium.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1.0 g/m <sup>3</sup> as CaCO <sub>3</sub> | 1         |
| Electrical Conductivity (EC)             | Conductivity meter, 25°C. Analysed at Hill Laboratories - Chemistry; 101c Waterloo Road, Christchurch. APHA 2510 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0.1 mS/m                                  | 1         |
| Electrical Conductivity (EC)             | Conductivity meter, 25°C. APHA 2510 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1 µS/cm                                   | 1         |
| Approx Total Dissolved Salts             | Calculation: from Electrical Conductivity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2 g/m <sup>3</sup>                        | 1         |
| Sample Temperature*                      | A nominal sample temperature of 20°C has been assumed by the laboratory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.1 °C                                    | 1         |
| Filtration for dissolved metals analysis | Sample filtration through 0.45µm membrane filter and preservation with nitric acid. APHA 3030 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                                         | 1         |
| Total Arsenic                            | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017 / US EPA 200.8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.0011 g/m <sup>3</sup>                   | 1         |
| Total Boron                              | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.0053 g/m <sup>3</sup>                   | 1         |
| Dissolved Calcium                        | Filtered sample, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.05 g/m <sup>3</sup>                     | 1         |
| Total Calcium                            | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.053 g/m <sup>3</sup>                    | 1         |

| Sample Type: Aqueous                |                                                                                                                                                                                                                                                                                                                            |                                           |           |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------|
| Test                                | Method Description                                                                                                                                                                                                                                                                                                         | Default Detection Limit                   | Sample No |
| Total Copper                        | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017 / US EPA 200.8.                                                                                                                                                                                                                          | 0.00053 g/m <sup>3</sup>                  | 1         |
| Total Iron                          | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                         | 0.021 g/m <sup>3</sup>                    | 1         |
| Total Lead                          | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017 / US EPA 200.8.                                                                                                                                                                                                                          | 0.00011 g/m <sup>3</sup>                  | 1         |
| Total Magnesium                     | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                         | 0.021 g/m <sup>3</sup>                    | 1         |
| Total Manganese                     | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017 / US EPA 200.8.                                                                                                                                                                                                                          | 0.00053 g/m <sup>3</sup>                  | 1         |
| Total Potassium                     | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                         | 0.053 g/m <sup>3</sup>                    | 1         |
| Total Sodium                        | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                         | 0.021 g/m <sup>3</sup>                    | 1         |
| Total Zinc                          | Nitric acid digestion, ICP-MS, trace level. APHA 3125 B 23 <sup>rd</sup> ed. 2017 / US EPA 200.8.                                                                                                                                                                                                                          | 0.0011 g/m <sup>3</sup>                   | 1         |
| Chloride                            | Filtered sample from Christchurch. Ion Chromatography. APHA 4110 B (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                   | 0.5 g/m <sup>3</sup>                      | 1         |
| Fluoride                            | Direct measurement, ion selective electrode. APHA 4500-F- C 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                                     | 0.05 g/m <sup>3</sup>                     | 1         |
| Total Ammoniacal-N                  | Filtered Sample from Christchurch. Phenol/hypochlorite colourimetry. Flow injection analyser. (NH <sub>4</sub> -N = NH <sub>4</sub> <sup>+</sup> -N + NH <sub>3</sub> -N). APHA 4500-NH <sub>3</sub> H (modified) 23 <sup>rd</sup> ed. 2017.                                                                               | 0.010 g/m <sup>3</sup>                    | 1         |
| Nitrate-N                           | Filtered sample from Christchurch. Ion Chromatography. APHA 4110 B (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                   | 0.05 g/m <sup>3</sup>                     | 1         |
| Total Phosphorus                    | Total phosphorus digestion, ascorbic acid colorimetry. Discrete Analyser. APHA 4500-P B & E (modified from manual analysis and also modified to include a reductant to reduce interference from any arsenic present in the sample) 23 <sup>rd</sup> ed. 2017. NWASCO, Water & soil Miscellaneous Publication No. 38, 1982. | 0.004 g/m <sup>3</sup>                    | 1         |
| Reactive Silica                     | Filtered sample. Heteropoly blue colorimetry. Flow Injection Analyser APHA 4500-SiO <sub>2</sub> F (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                   | 0.10 g/m <sup>3</sup> as SiO <sub>2</sub> | 1         |
| Sulphate                            | Filtered sample from Christchurch. Ion Chromatography. APHA 4110 B (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                                                                                   | 0.5 g/m <sup>3</sup>                      | 1         |
| Non-Purgeable Organic Carbon (NPOC) | Acidification, purging to remove inorganic C, super-critical persulphate oxidation at 375°C, IR detection. APHA 5310 C (modified) 23 <sup>rd</sup> ed. 2017.                                                                                                                                                               | 0.3 g/m <sup>3</sup>                      | 1         |
| Routine Water Profile               |                                                                                                                                                                                                                                                                                                                            | -                                         | 1         |
| Total Coliforms and E.coli          |                                                                                                                                                                                                                                                                                                                            |                                           |           |
| Total Coliforms                     | MPN count using Colilert 18 (Incubated at 35°C for 18 hours) and 97 wells. Analysed at Hill Laboratories - Microbiology; 101c Waterloo Road, Hornby, Christchurch. APHA 9223 B 23 <sup>rd</sup> ed. 2017.                                                                                                                  | 1 MPN / 100mL                             | 1         |
| Escherichia coli                    | MPN count using Colilert 18 (Incubated at 35°C for 18 hours) and 97 wells. Analysed at Hill Laboratories - Microbiology; 101c Waterloo Road, Hornby, Christchurch. APHA 9223 B 23 <sup>rd</sup> ed. 2017.                                                                                                                  | 1 MPN / 100mL                             | 1         |

These samples were collected by yourselves (or your agent) and analysed as received at the laboratory.

Testing was completed between 21-Nov-2020 and 26-Nov-2020. For completion dates of individual analyses please contact the laboratory.

Samples are held at the laboratory after reporting for a length of time based on the stability of the samples and analytes being tested (considering any preservation used), and the storage space available. Once the storage period is completed, the samples are discarded unless otherwise agreed with the customer. Extended storage times may incur additional charges.

This certificate of analysis must not be reproduced, except in full, without the written consent of the signatory.



Ara Heron BSc (Tech)  
Client Services Manager - Environmental